more specific to our task (human faces): mid-level features contain combinations of shapes that form objects in the human face like eyes and noses. As we go deeper through the network, we notice that features eventually transition from general to specific and, by the last layer of the network, form high-level features that are very specific to our task. We start seeing parts of human faces that distinguish one person from another.

Now, let's take this example and compare the feature maps extracted from four models that are trained to classify faces, cars, elephants, and chairs (see figure 6.7). Notice that the earlier layers' features are very similar for all the models. They represent low-level features like edges, lines, and blobs. This means models that are trained on one task capture similar relations in the data types in the earlier layers of the network and can easily be reused for different problems in other domains. The deeper we go into the network, the more specific the features, until the network overfits its training data and it becomes harder to generalize to different tasks. The lower-level features are almost always transferable from one task to another because they contain generic information like the structure and nature of how images look. Transferring information like lines, dots, curves, and small parts of objects is very valuable for the network to learn faster and with less data on the new task.

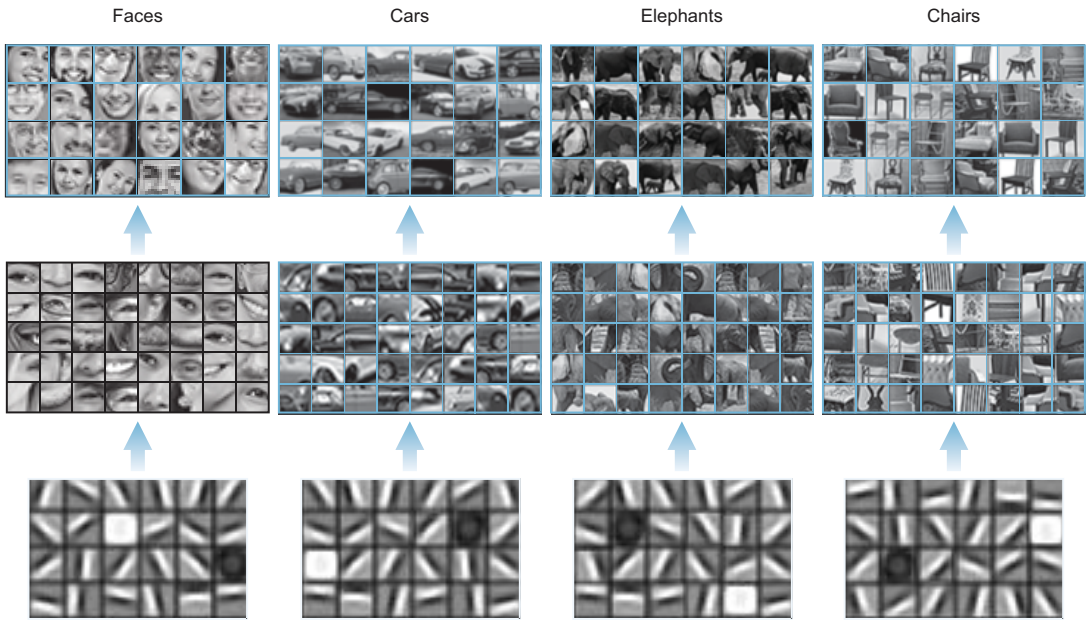

Figure 6.7 Feature maps extracted from four models that are trained to classify faces, cars, elephants, and chairs